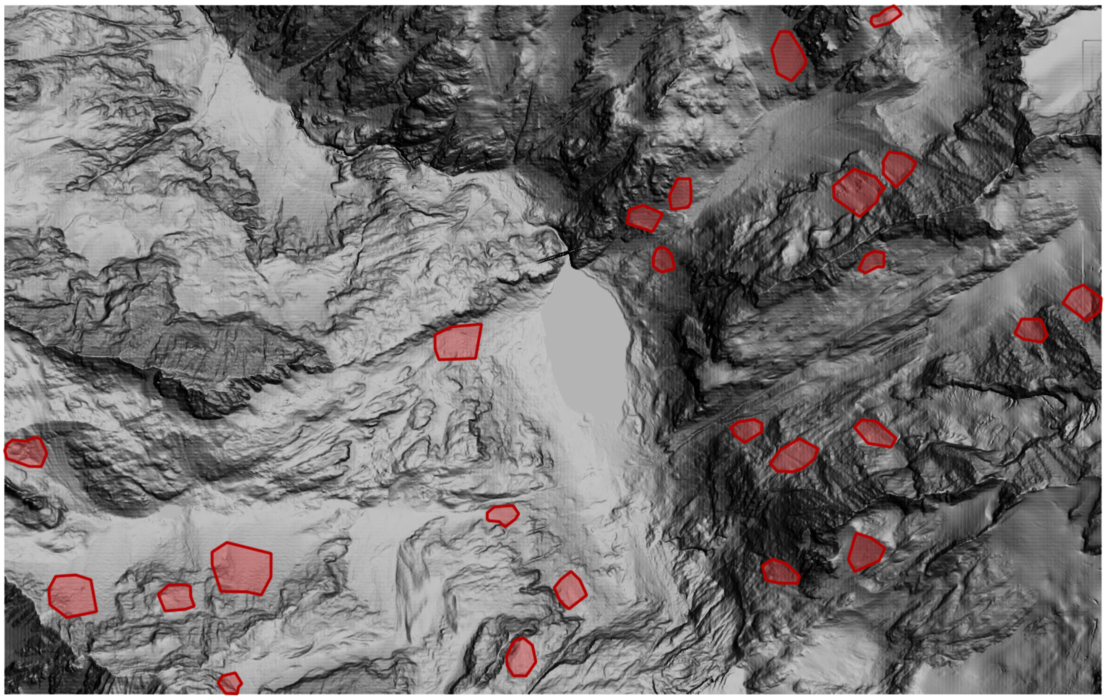
AVAC4QGIS Tutorial

From an avalanche release to an optional lake impulse wave

LHE–EPFL, Switzerland

August 28, 2026

This tutorial follows the interface distributed with AVAC4QGIS 0.5.14. It uses the public demonstration terrain and avalanche-release layer supplied in the repository. The values visible in screenshots belong to one demonstration workspace; they are not recommended values for another site. Choose scientific parameters from the data and objectives of the case being studied.



The tutorial terrain DEM displayed as hillshade, with the release polygons in red.

Contents

Part I

General workflow

1 What AVAC4QGIS simulates

AVAC4QGIS brings two separate depth-averaged solvers into one QGIS workflow:

- **AVAC** simulates the release, acceleration, propagation, and deposition of an avalanche over a terrain DEM.
- **WAVE** simulates the water response in a lake or reservoir. It is optional and is enabled only when the avalanche-to-water interaction is part of the case.

An AVAC-only study ends after the avalanche results are produced. In a coupled study, WAVE starts from a completed AVAC run and converts the avalanche flux entering the lake through the shoreline into internal water mass and momentum sources. WAVE then propagates the resulting impulse wave. Enabling WAVE does not change or rerun AVAC.

2 Tutorial data and preparation

Load the following two solver inputs from the repository's `tutorial/` directory into the QGIS project:

File	Purpose
`topo_cut_dam_1m.tif`	Terrain elevation raster used by AVAC and, in the coupled example, by WAVE.
`avalanches.geojson`	Polygon layer defining the possible avalanche releases.

The optional georeferenced image `satellite.png`, together with its `satellite.pgw` world file, is only a visual background. It is not a solver input.

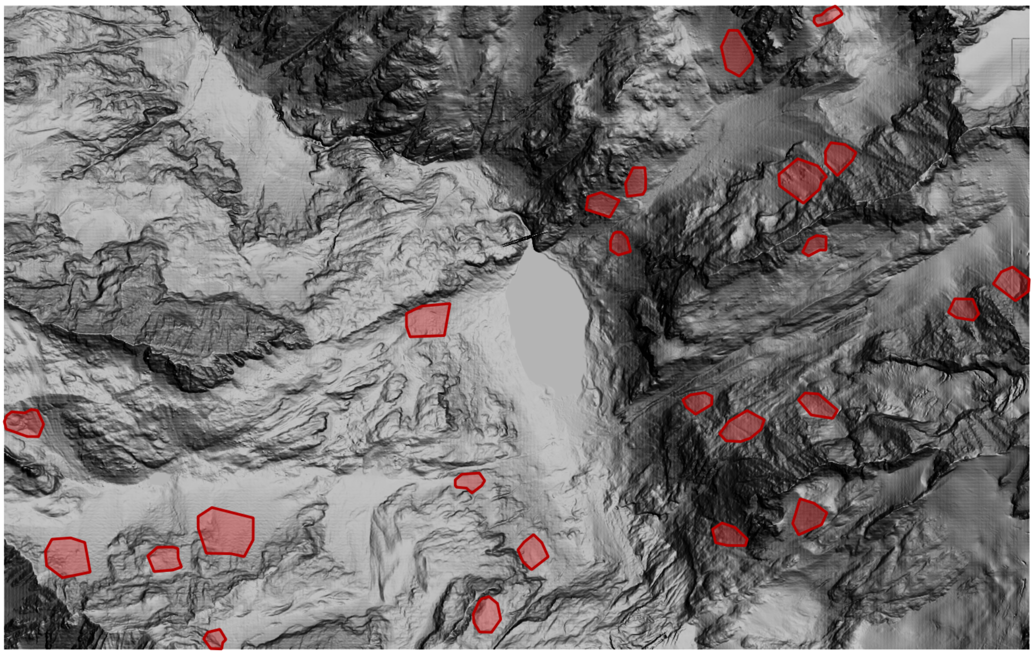


Figure 1: The two required tutorial layers: the DEM displayed with a hillshade renderer and the release-polygon layer.

3 The persistent controls above the tabs

Open the plugin from **Plugins** > **AVAC4QGIS**. The controls above the tabs apply to the complete plugin workflow:

1. Set the **AVAC Working Directory** to a writable case folder. Prepared runs, result rasters, previews, and exports are stored there.

2. Leave **Enable Lake-Wave Extension** off for an avalanche-only case. Turn it on to show the WAVE Parameters and WAVE Run tabs.

3. **Save Case** records the complete plugin setup, including the working directory, persistent input-layer paths, visible AVAC settings, and WAVE settings when the extension is enabled.

4. **Load Case** restores that complete setup after QGIS or the plugin has been closed.

Load Case/**Save Case** concern the complete plugin. The configuration buttons at the bottom of AVAC Parameters and WAVE Parameters concern only the corresponding solver settings. A QGIS memory layer must be saved to disk before it can be restored portably from a case file.

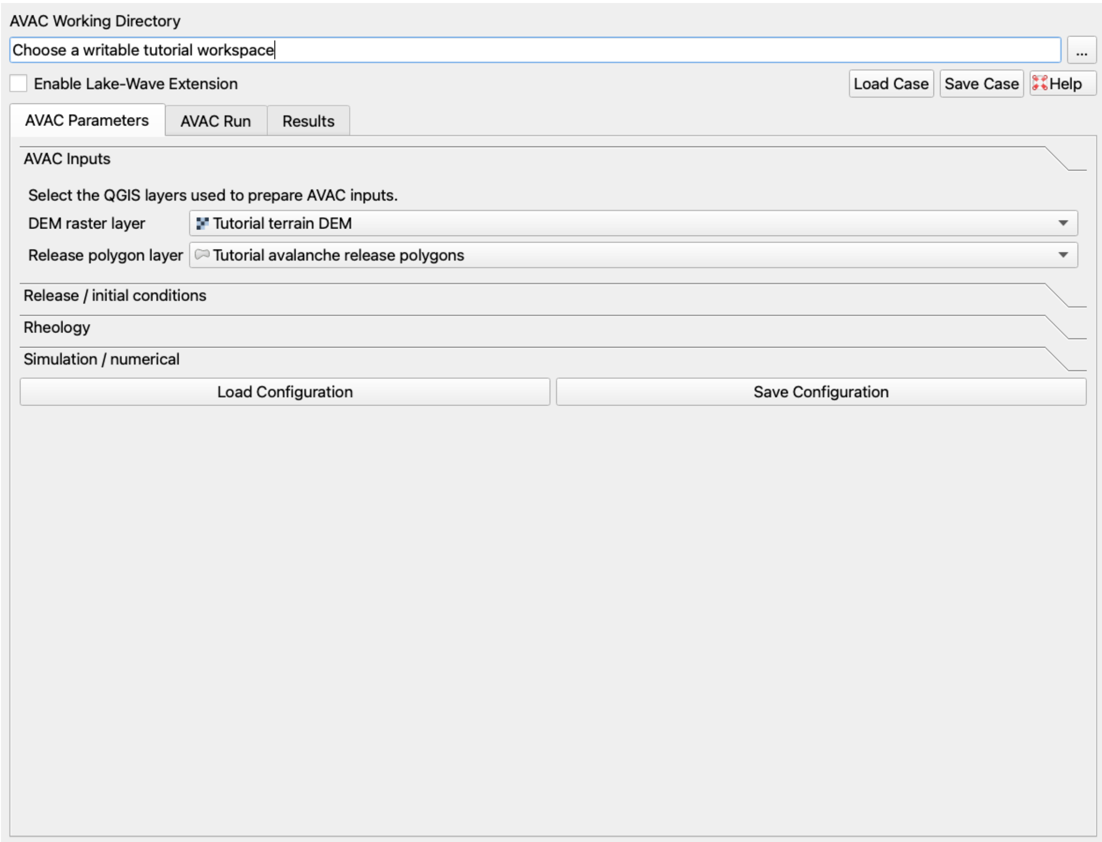


Figure 2: The complete AVAC4QGIS dock. Persistent case controls remain above every workflow tab.

Part II

AVAC: preparing and running the avalanche

4 Minimum inputs and AVAC configuration files

Only two map inputs are required for a new AVAC simulation: a terrain DEM and a release polygon. The four accordions in AVAC Parameters collect those inputs and the release, rheology, and numerical settings.

At the bottom of the tab, **Load Configuration** reads AVAC settings from an AVAC YAML file and **Save Configuration** writes the current AVAC settings. These buttons are useful for transferring the avalanche model setup without replacing the rest of a saved plugin case. Load a configuration before reviewing the four accordions, because loaded values replace their current contents.

5 AVAC Inputs accordion

1. Select **Tutorial terrain DEM** as the **DEM raster layer**.
2. Select **Tutorial avalanche release polygons** as the **Release polygon layer**.
3. If the layer contains several features, select the releases to be simulated in the QGIS map. If no feature is selected, all usable release features are used.

The DEM must have a valid projected CRS, finite usable cells, and square pixels. The release layer must be a valid polygon or multipolygon layer with a valid CRS. At least one selected release must overlap a usable DEM cell. AVAC uses the full usable DEM extent.

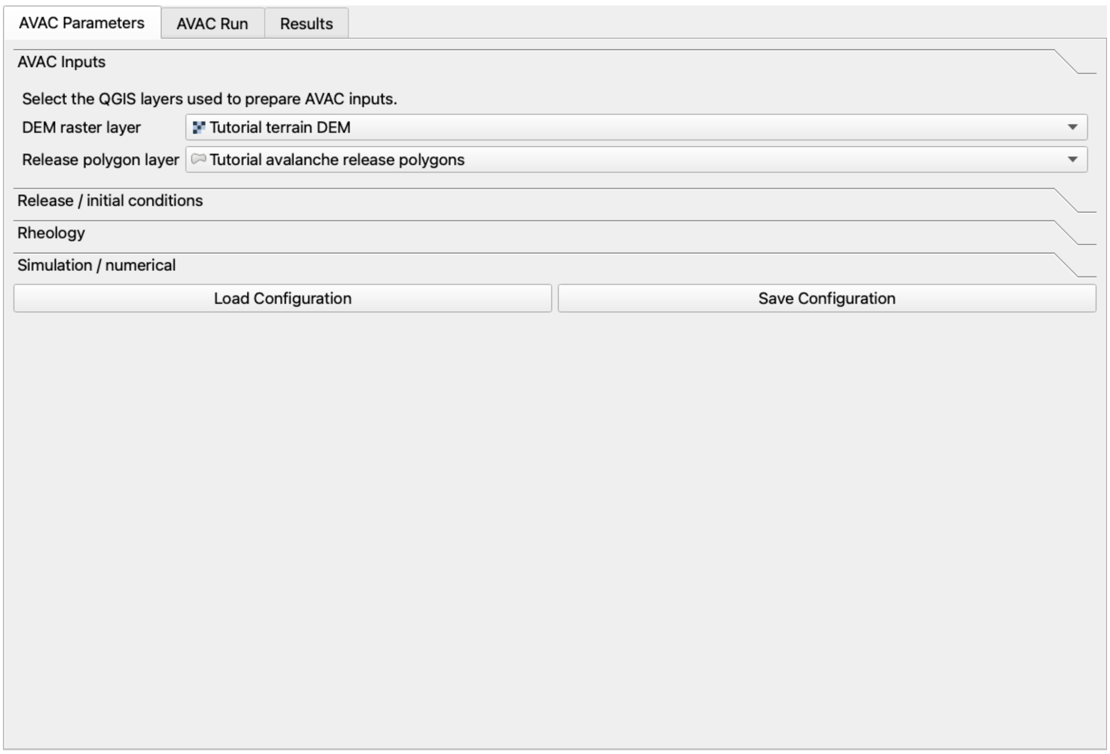


Figure 3: AVAC Inputs accordion with the tutorial DEM and release layer selected.

6 Release / initial conditions accordion

This page converts release polygons into the initial mobile avalanche depth. Choose the release-depth interpretation, the reference terrain or slope quantity used by that interpretation, and any volume or depth scaling required by the selected method. The initial AVAC depth is zero outside the release polygons: the model does not silently add a stationary snow cover.

Use **Preview Initial Depth** before preparation. The preview writes a raster in the working directory and loads it in QGIS, allowing the selected release features, rasterization, and resulting initial volume to be checked.

> The release layer represents the initial moving volume. Its polygons must not be confused with a computational mask or with the later lake polygon.

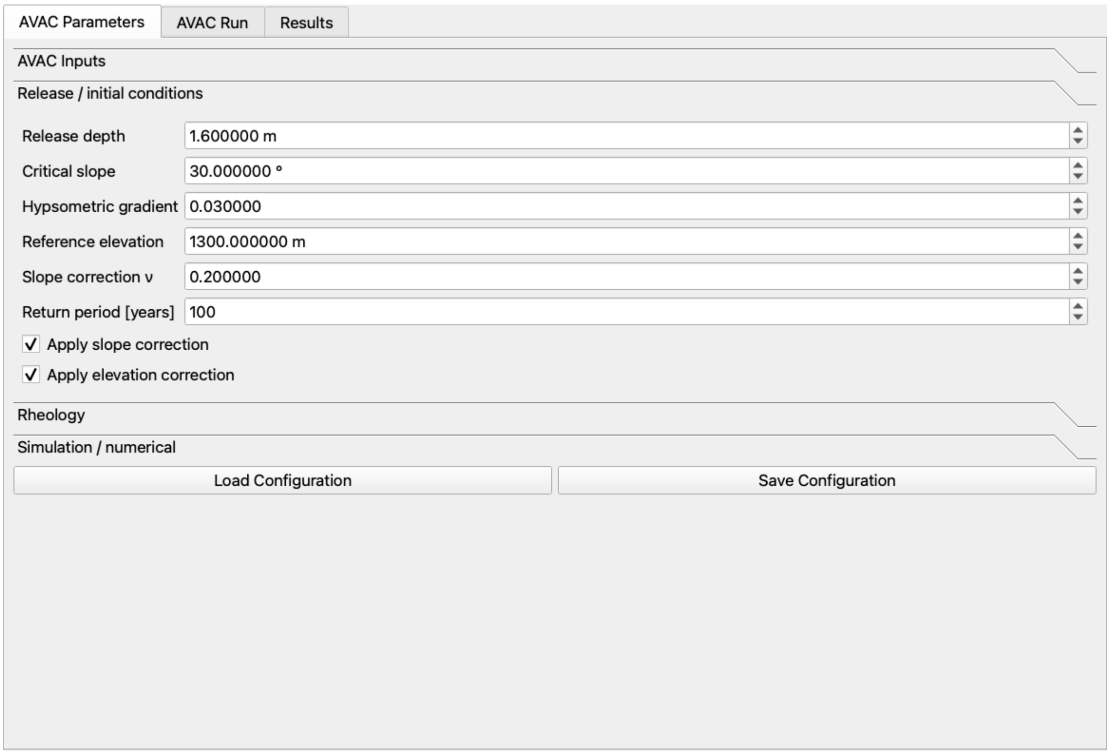


Figure 4: Release / initial conditions accordion. Review the method, its inputs, and the preview control.

Figure 5 shows the spatial check produced for the tutorial's uniform-depth release. Colored cells are the initial mobile depth that AVAC will place inside the selected polygons; the transparent background confirms that no depth was created elsewhere.

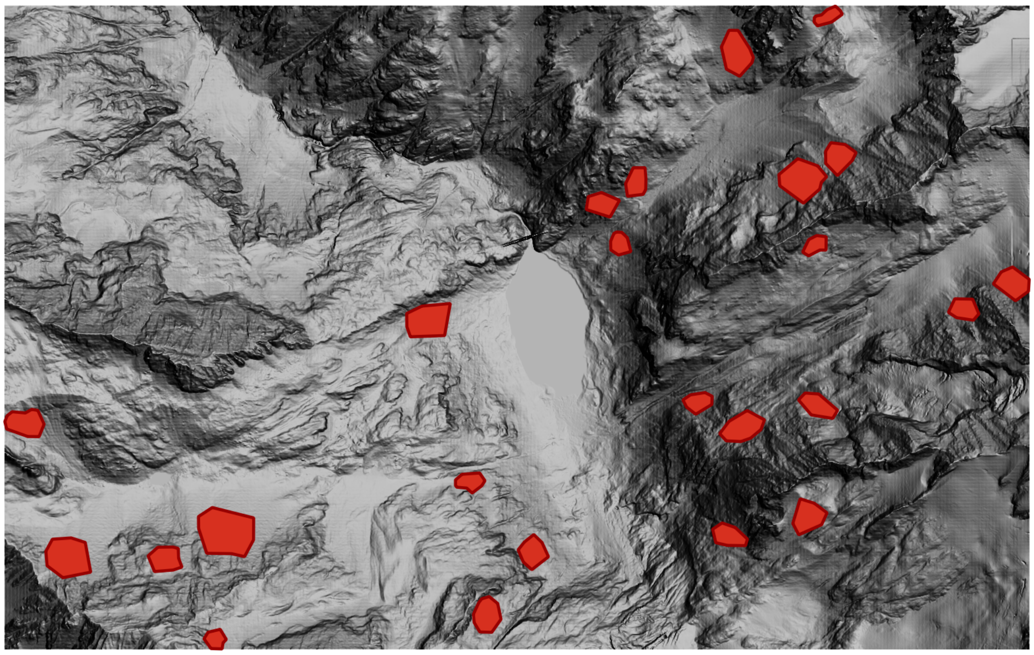


Figure 5: Preview Initial Depth output for the tutorial release polygons.

7 Rheology accordion

Choose the rheological model and enter its physical parameters. The visible fields describe resistance to motion and, when enabled, how those values vary with elevation. The interface changes which fields are active according to the selected model.

Use **Visualize Rheology Zones** to create a categorical raster from the current elevation breaks. Each cell color identifies the zone whose parameters the solver will apply there. This is a setup check, not a simulation result.

How to enter altitude zones

Leave **Altitude zones** empty to use the single μ, ξ, and cohesion values shown above it everywhere. For altitude-dependent rheology, enter one comma-separated row per zone in this exact order:

```
μ, ξ, cohesion (Pa), lower elevation (m)
```

The first row is the lowest zone and its fourth field must remain blank. Every following row begins at the lower elevation written in its fourth field; those elevations must be strictly increasing. For example:

```
0.30,   600,   0,
0.225,  1200,  0,  1680
```

This applies $\mu = 0.30$ and $\xi = 600$ m/s^2 below 1680 m, and $\mu = 0.225$ and $\xi = 1200$ m/s^2 at and above 1680 m. Cohesion is zero in both rows. The solver uses ξ for the Voellmy variants and cohesion only for cohesive Voellmy; the Coulomb model uses μ. All four columns are still required so that the zone definition has one unambiguous format.

Check the categorical raster whenever altitude-dependent values are used. Every usable DEM cell must receive the expected zone, including cells above the highest breakpoint. Parameter meaning and units are summarized in the separate User Interface Reference.

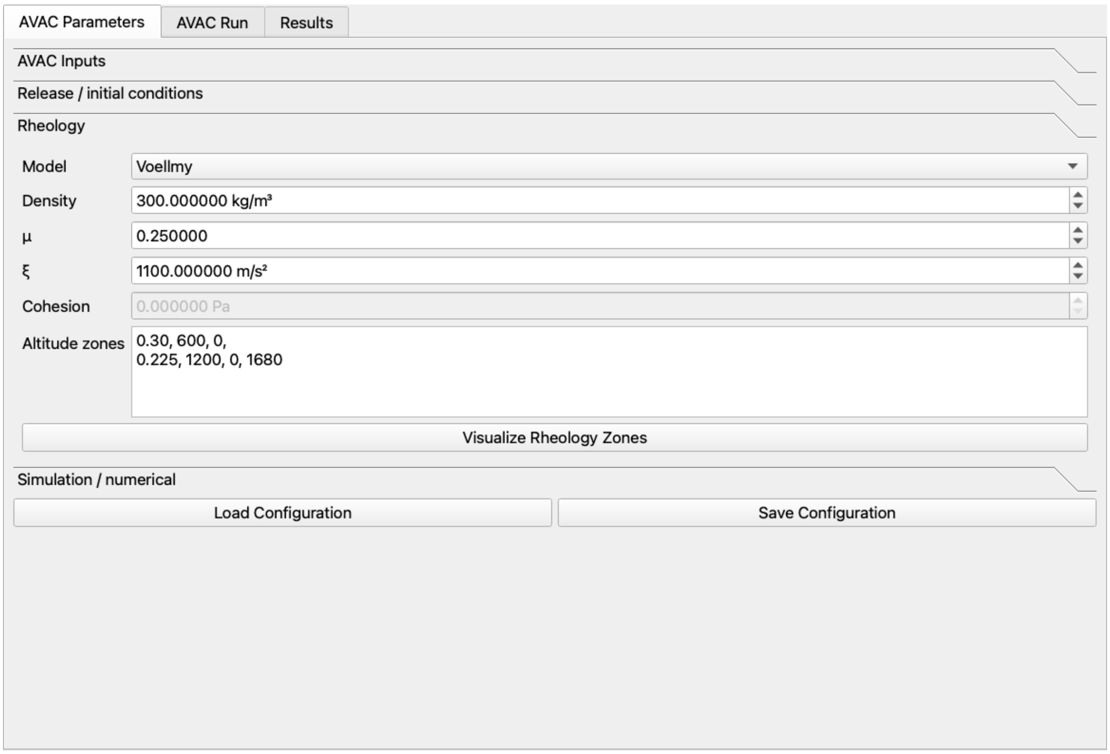


Figure 6: Rheology accordion with the model controls and rheology-zone visualization button.

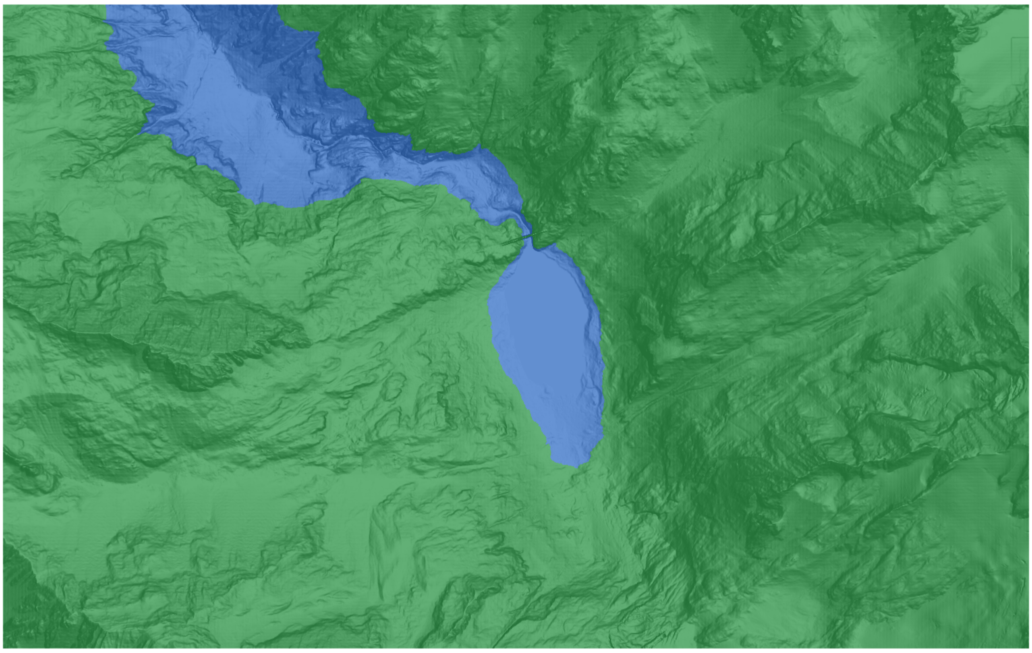


Figure 7: Visualize Rheology Zones output for the two-zone example. Blue identifies elevations below 1680 m and green identifies elevations at or above 1680 m; the hillshade remains visible beneath the categorical raster.

8 Simulation / numerical accordion

Set the avalanche duration, solver-grid cell size, CFL controls, dry-depth tolerance, refinement level, and limiter. The current interface uses an open outer boundary: material leaving the DEM calculation domain is allowed to leave and is not reflected into the domain.

The AVAC solver cell size must equal the DEM pixel size or be a whole-number multiple of it. Both coordinate spans must contain whole solver cells. A coarser cell reduces cost but also reduces spatial detail. CFL limits must be internally consistent, and increasing AMR or core count does not replace the need for a resolution study.

AVAC outputs are written at the fixed interval used by the packaged workflow. The number of output frames follows from that interval and the requested duration.

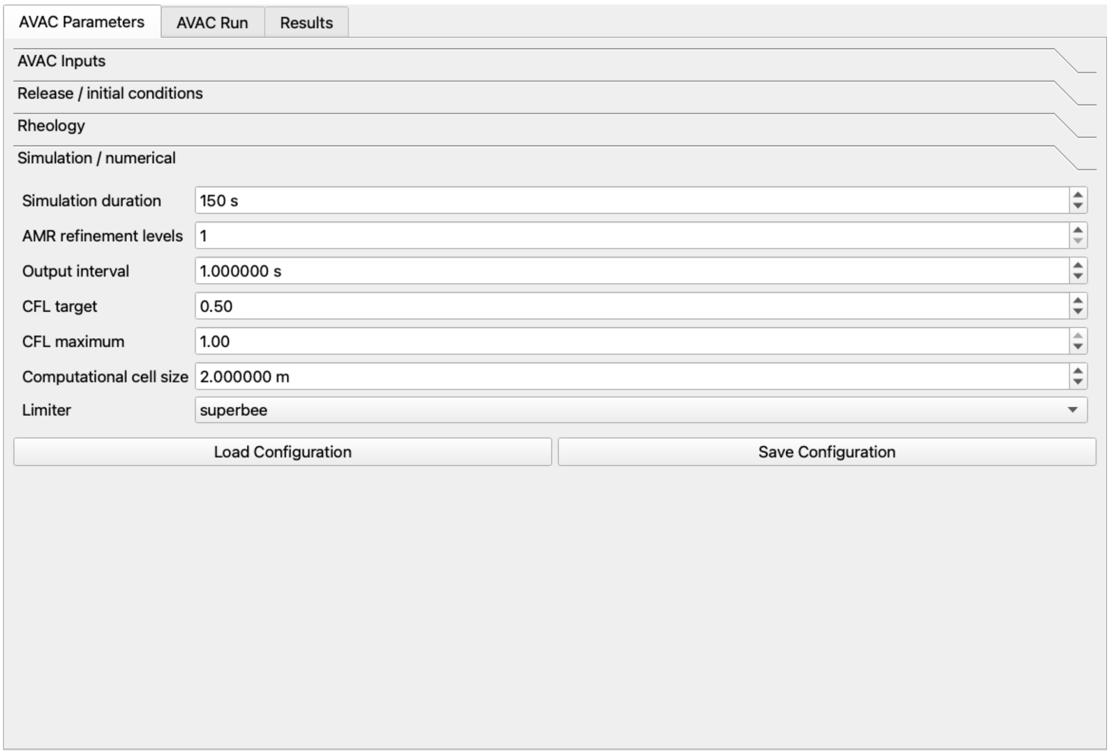


Figure 8: Simulation / numerical accordion. The screenshot shows every numerical control exposed to the user.

9 Check, prepare, and run AVAC

Open the **AVAC Run** tab and work from top to bottom:

1. Set **CPU cores**. The initial value is the number available on the machine; reduce it if other applications need resources.
2. Select **Check Environment**. A packaged-runtime installation does not require a system compiler or a separate Clawpack installation.
3. Select **Validate Inputs**. Correct every reported preflight failure before continuing.
4. Optionally rerun **Preview Initial Depth** and **Visualize Rheology Zones**.
5. Select **Prepare AVAC Run**. A new, isolated run folder is created and the progress bar reports input preparation.

6. Select **Run**. The solver runs asynchronously; the same progress area reports simulation progress. **Stop** requests a controlled termination if the run must be canceled.

7. Use **Show AVAC Log** when diagnostic details are needed. A successful run is automatically listed in Results.

Preparation and execution are separate. If any AVAC input or parameter is changed after preparation, prepare a new run before pressing Run.

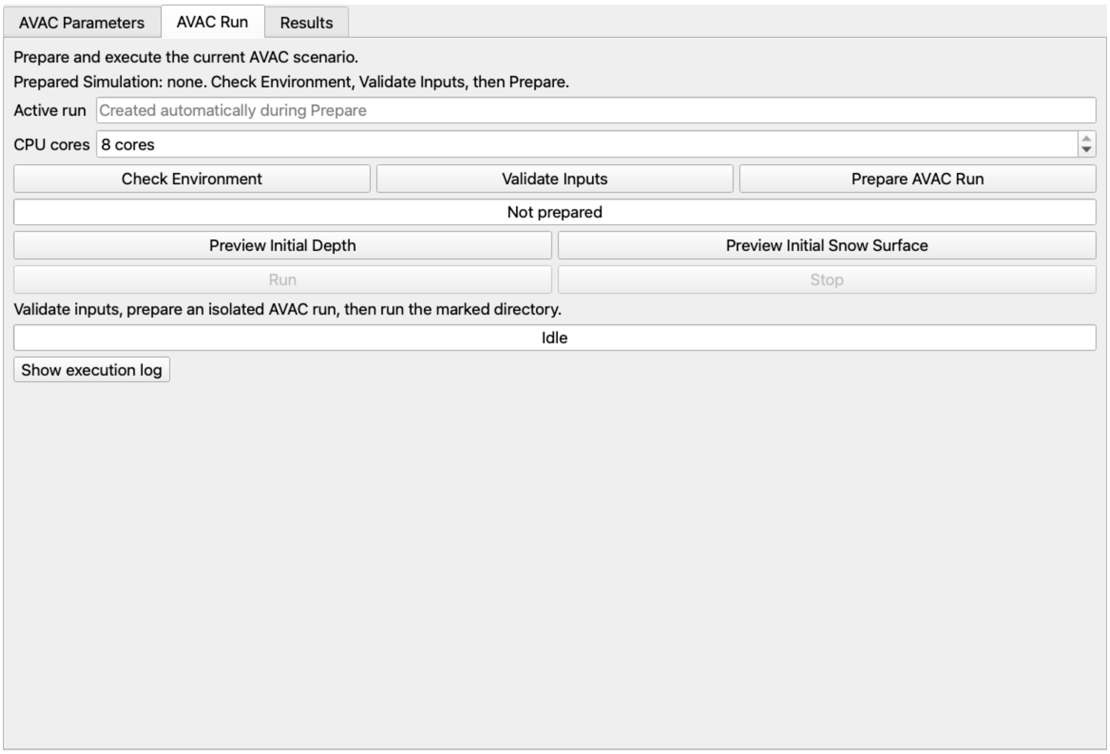


Figure 9: AVAC Run tab. Environment checks, validation, preparation, execution, progress, and the optional log are grouped here.

Part III

WAVE: optional avalanche–lake coupling

10 Prerequisite and coupling concept

Enable the Lake-Wave Extension only after a completed AVAC run with usable time-dependent results exists. WAVE reads that run without modifying it and inherits its rectangular calculation bounds, simulation duration, output times, and temporal origin.

The lake polygon has two distinct roles:

- it defines which WAVE cells are initially part of the water body at the selected absolute water-surface elevation; and
- it defines the internal shoreline where the entering avalanche is converted into water mass and momentum sources.

At every AVAC output time, the coupling samples avalanche depth and both horizontal momentum components along the grid-aligned WAVE shoreline. Only the flux directed from land into the initially wet lake is admitted. The integrated inward volume flux and its two momentum components are then added to the WAVE cells adjacent to that shoreline. Cells outside the initial lake are not permanently forced dry, so runup can wet terrain beyond the polygon.

> The polygon is not the WAVE calculation domain. It separates the initial water body from surrounding terrain and locates the internal avalanche-to-water transfer.

11 AVAC Source accordion

Choose the completed AVAC run whose arrival at the lake will generate the impulse wave. The selector lists completed runs in the working directory; use **Refresh** after a run finishes. The summary below the selector helps confirm the intended source, duration, and output availability.

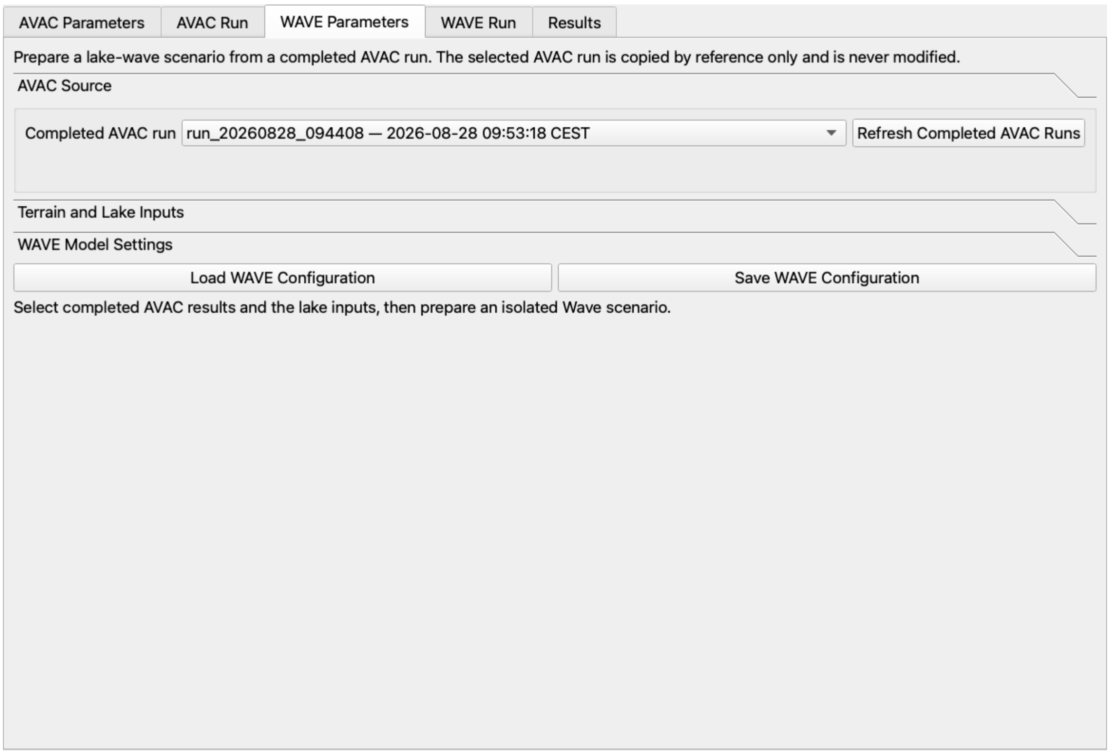


Figure 10: AVAC Source accordion. WAVE preparation always begins from a completed AVAC run.

12 Terrain and Lake Inputs accordion

1. Select the terrain DEM. It must cover the AVAC calculation bounds.
2. Select an existing **Lake / water-body polygon**, or create one from a water-surface elevation and a clicked point in the DEM-connected basin.
3. Enter the **Water level**. This is an absolute elevation in the same vertical datum as the DEM, not a water depth.
4. Enter the WAVE grid cell size and select **Preview Water Level**. The preview creates the exact solver-grid initial lake state. If these inputs remain unchanged, WAVE preparation reuses the cached preview rather than calculating it again.

To create the water body without a preexisting polygon, first select the DEM and enter the absolute water level. Select **Create Lake Polygon from Map Point**; the cursor is armed for one click. Click a DEM cell inside the intended basin and below the chosen level. The plugin transforms the click to the DEM CRS and flood-fills only the north–south/east–west connected DEM cells at or below that elevation. Other low areas that are not connected to the clicked basin are excluded. The connected mask is converted to a GeoPackage in `derived_inputs/`, loaded into QGIS, and selected as the lake polygon. A right click cancels point capture.

The connected contour must close inside the terrain DEM. If it reaches a DEM edge, use a DEM that fully encloses the water body or reconsider the water level. After the polygon is created, select **Preview Water Level**. This resamples the terrain and polygon onto the chosen WAVE grid, sets the initial water surface to the absolute level, and displays water depth ($h = \max(0, \eta_0 - z)$) in the connected lake cells. Inspecting this exact solver-grid preview catches shoreline differences introduced by a coarser WAVE cell size before preparation.

The WAVE DEM must have a valid projected CRS and square pixels. Its cell size must equal the DEM resolution or be a whole-number multiple of it, and each inherited AVAC-domain span must contain at least two whole WAVE cells. The lake polygon and DEM must overlap. A zero water-surface elevation does not mean "empty lake"; it is interpreted as an absolute elevation in the DEM datum.

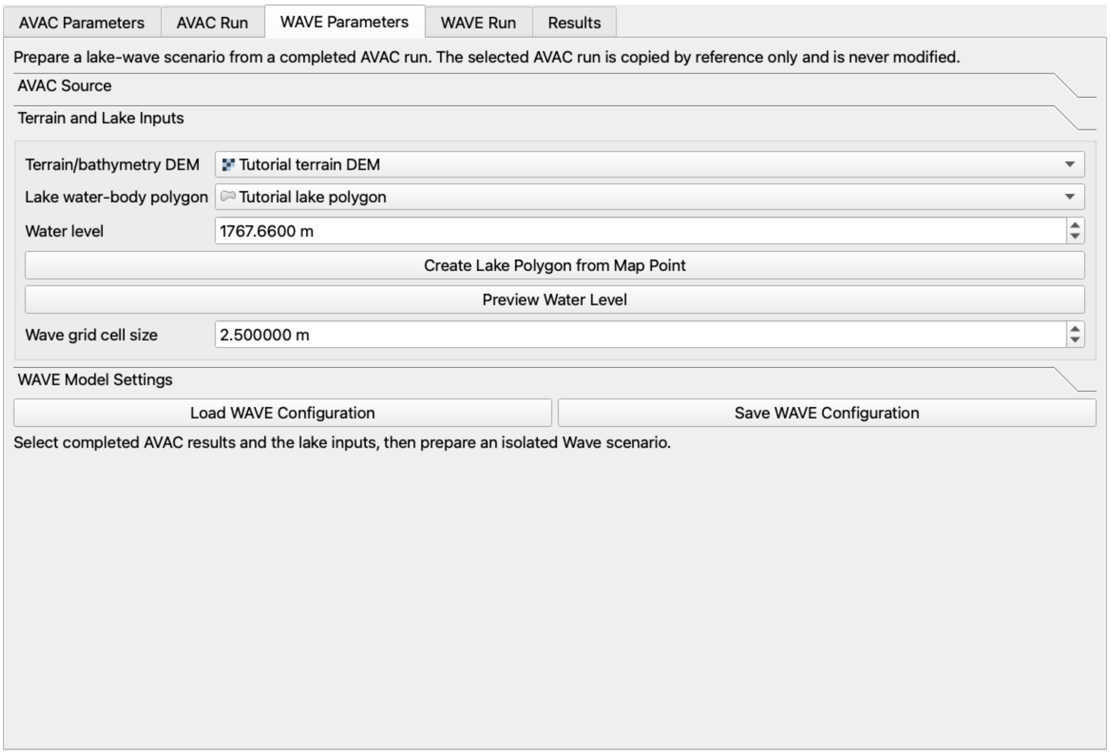
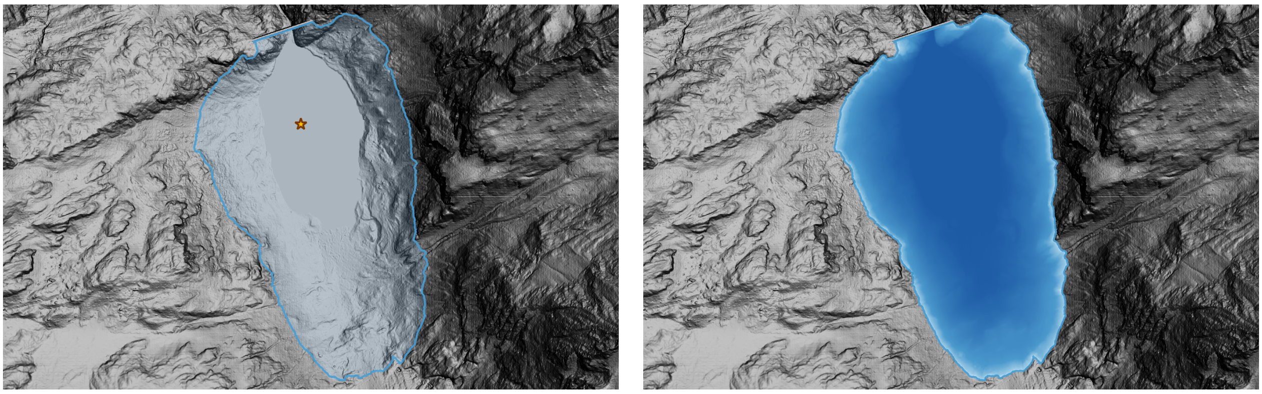


Figure 11: Terrain and Lake Inputs accordion, including the two lake-polygon paths and exact water-level preview.



(a) Clicked seed (gold star) and the connected water-body polygon.

(b) Exact WAVE-grid initial water-depth preview.

Figure 12: Building and checking the lake. The light-blue outline is the derived connected water-body polygon.

13 WAVE Model Settings accordion

Review the dry-depth threshold, CFL controls, damping and friction settings, refinement, limiter, and maximum solver steps. These values govern wet/dry behavior, numerical stability, energy loss, bed resistance, and computational cost. Use physically justified values and keep the CFL target below its maximum.

At the bottom of WAVE Parameters, **Load WAVE Configuration** and **Save WAVE Configuration** transfer only the WAVE setup. They do not replace the selected AVAC configuration or the complete plugin case.

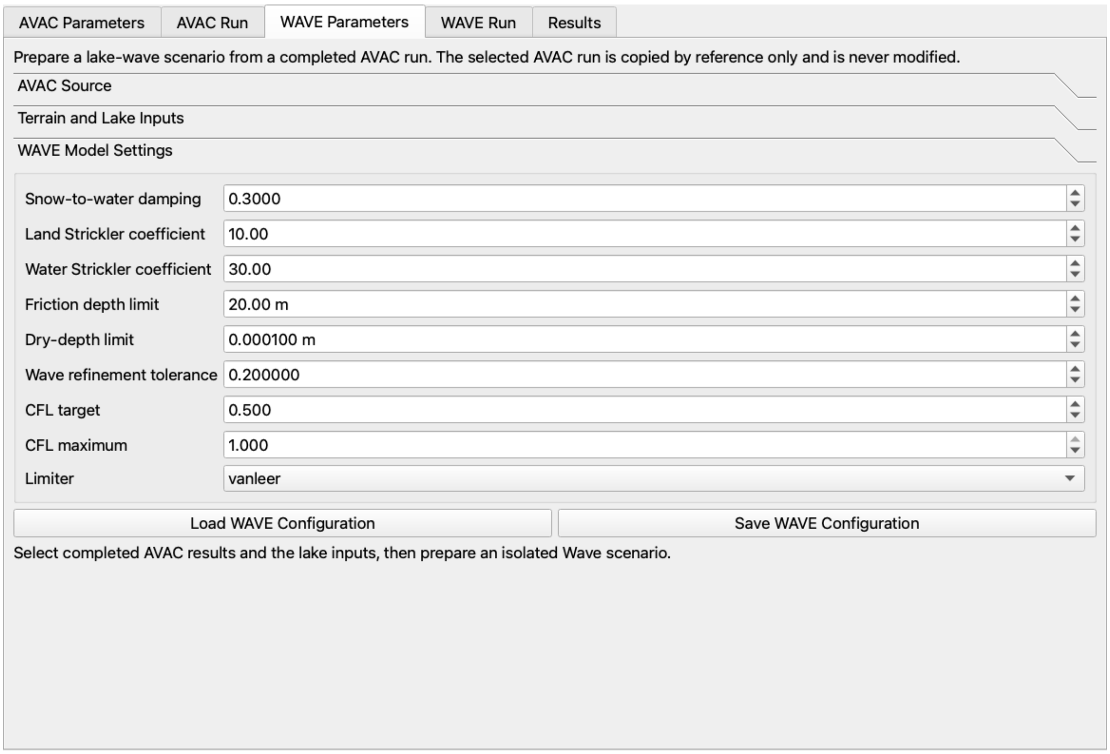


Figure 13: WAVE Model Settings accordion and the WAVE-only configuration controls.

14 Check, prepare, and run WAVE

Open **WAVE Run** and follow the same execution pattern as AVAC:

1. Set the desired **CPU cores** and select **Check Environment**.
2. Select **Validate Inputs**. WAVE verifies the completed AVAC source, terrain, polygon, preview state, inherited domain, and solver settings.
3. Select **Prepare WAVE Run**. Preparation builds a new WAVE scenario, the initial water state, and the internal shoreline source. If no avalanche mass crosses into the lake, preparation stops with a clear message because there is no impulse wave to simulate.
4. Select **Run**. Watch the progress bar, use **Stop** if necessary, and enable **Show WAVE Log** only when detailed output is useful.

The preparation summary records transferred volume and both horizontal momentum components. This ledger is the first coupling diagnostic to inspect when the simulated wave appears too small, too large, or directed incorrectly.

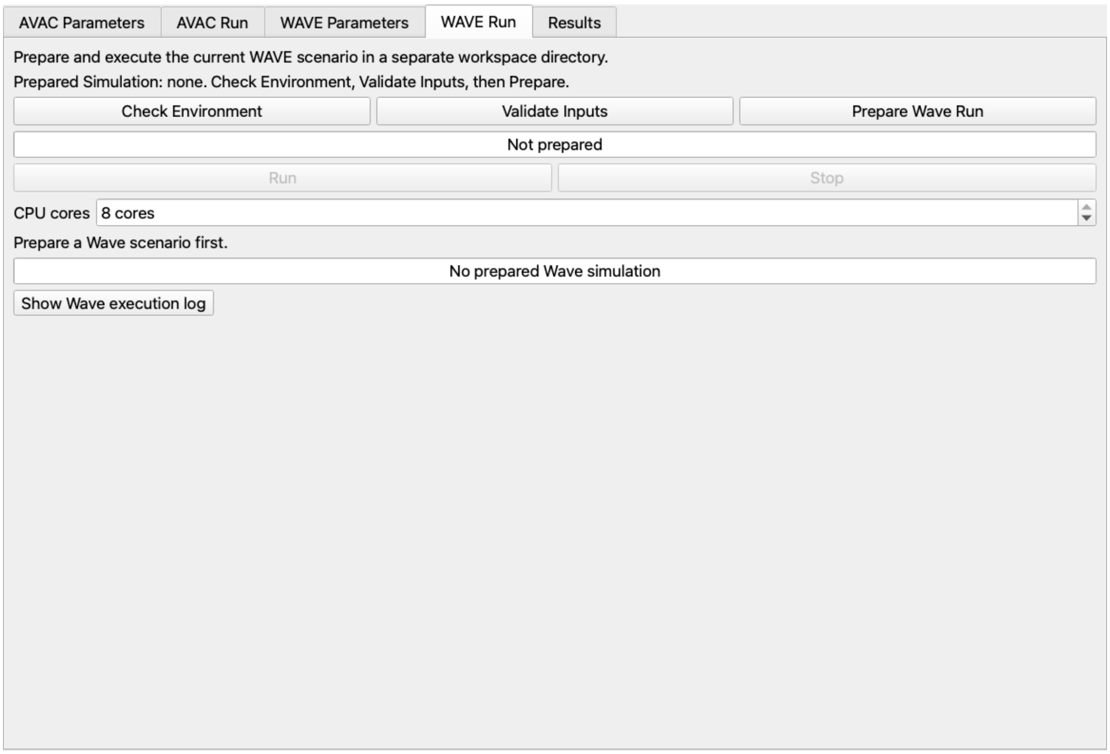


Figure 14: WAVE Run tab. It mirrors the AVAC check–validate–prepare–run sequence.

Part IV

Results: maps, animations, profiles, and histories

15 Select completed results

The final Results tab is shared by AVAC and WAVE. When WAVE is disabled, WAVE controls are hidden. When it is enabled, AVAC and WAVE result selectors appear together and all compatible variables use one temporal origin.

In the **Results Run** accordion, select a completed AVAC run and, when needed, its completed WAVE scenario. Use the refresh buttons if a run has just finished. The summaries identify what is selected; loading a new result does not alter solver output.

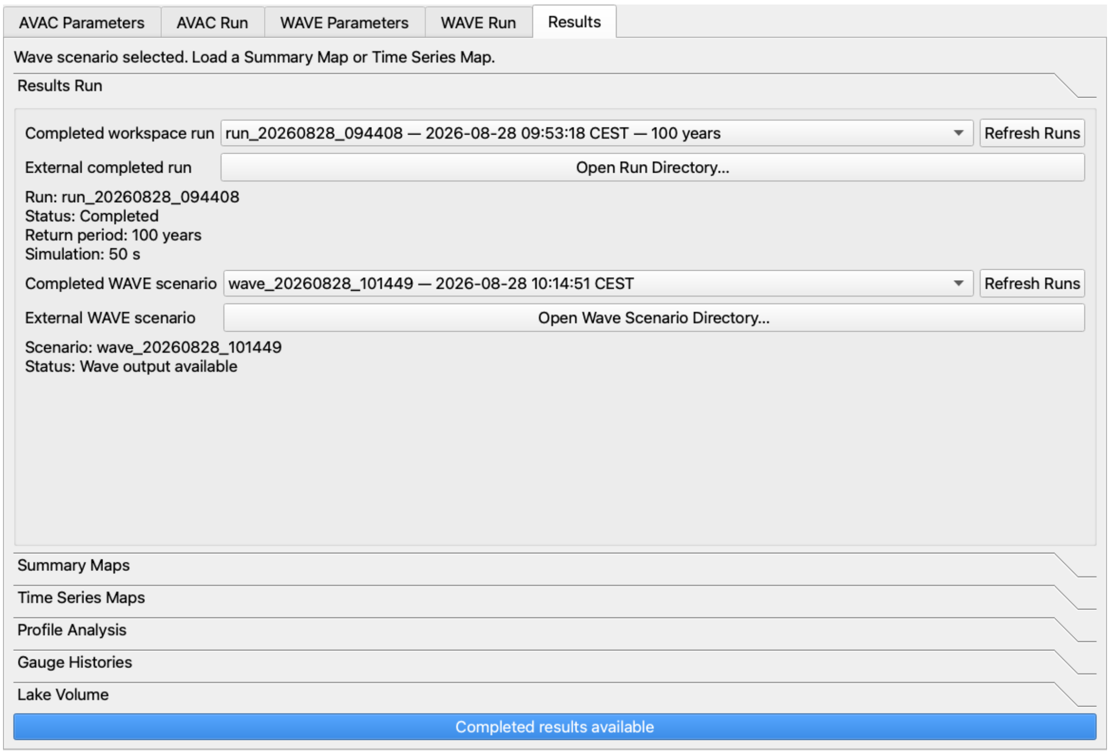


Figure 15: Results Run accordion with completed AVAC and WAVE sources selected together.

16 Summary Maps accordion

Summary maps collapse the complete simulation into one raster. AVAC provides peak snow depth, velocity, dynamic pressure, arrival time, and the time of selected maxima. WAVE provides maximum water-surface rise and maximum drawdown. Choose a map and select **Load Summary Map**.

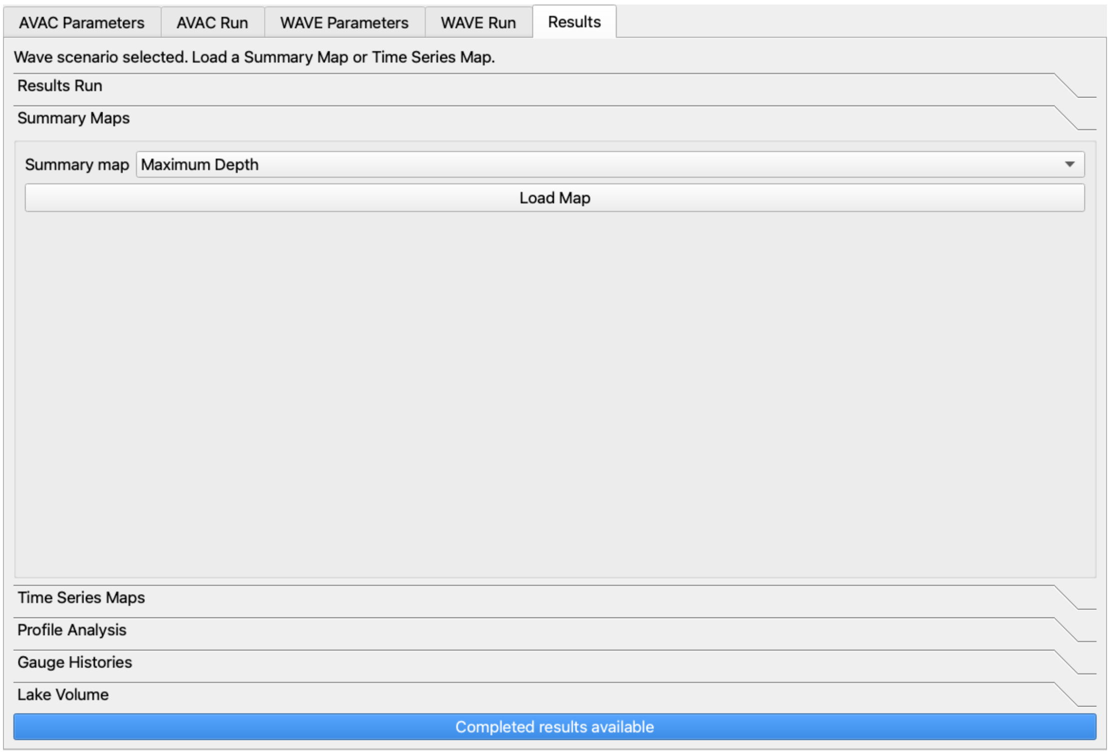
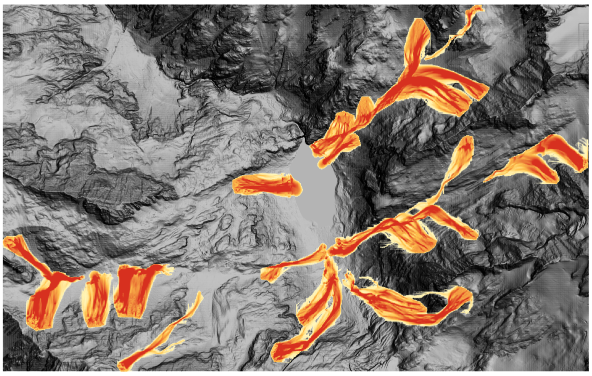
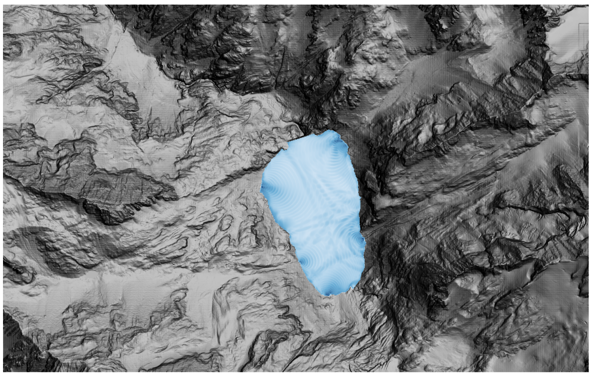


Figure 16: Summary Maps accordion. AVAC and WAVE products are selected from the same control.



(a) AVAC peak avalanche depth.



(b) WAVE maximum surface rise.

Figure 17: Examples of solver-wide summary rasters. Transparent zero values keep the terrain visible.

17 Time Series Maps accordion

Temporal maps retain one raster band per simulation frame. AVAC variables include snow depth, velocity, pressure, and snow-surface elevation. WAVE variables include water depth, water elevation, surface displacement, and the derived avalanche depth outside the lake. Load a variable, then use the frame slider, play/pause controls, or the PNG export tools.

The PNG export can write the current frame or the full series, uses the common simulation timeline, and can include legends and a scale bar. Full-series exports report progress and can be canceled. A missing temporal layer needed by another Results tool is loaded automatically.

> For a coupled visualization, use **WAVE Snow Depth Outside Lake** together with **WAVE Surface Displacement**. The first is the AVAC depth with the initial lake-polygon area set to zero; the second shows water rise and drawdown with zero displacement transparent. The original AVAC result is not modified.

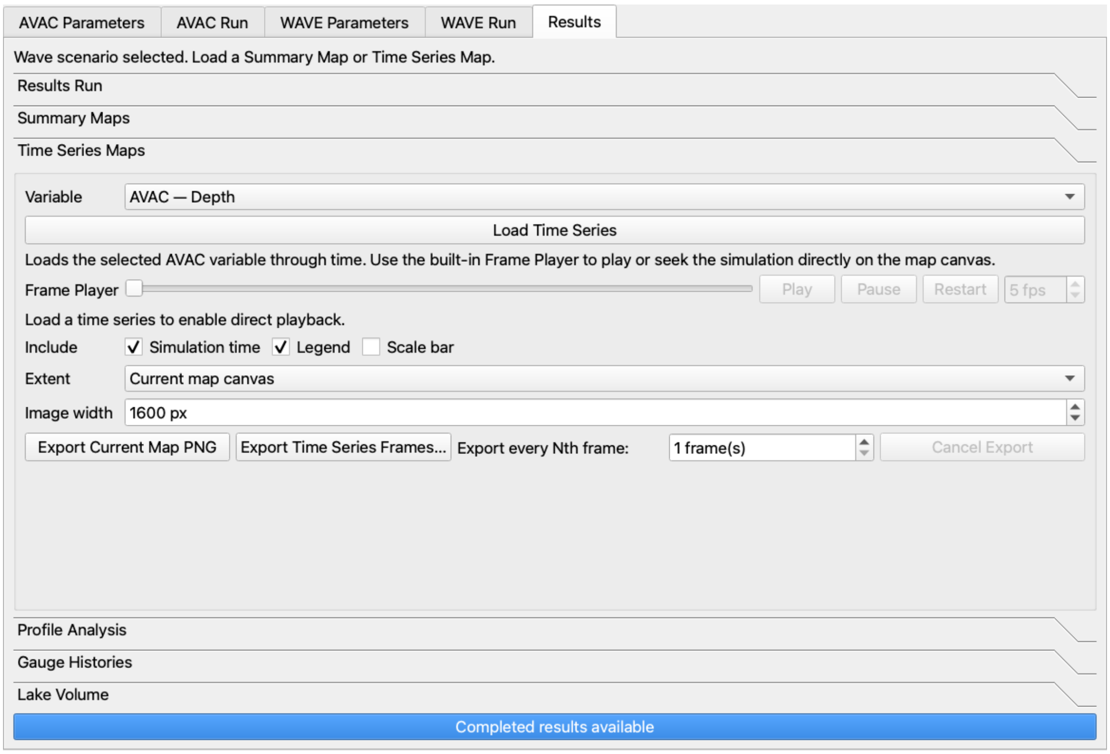


Figure 18: Time Series Maps accordion with playback, layer loading, and PNG export controls.

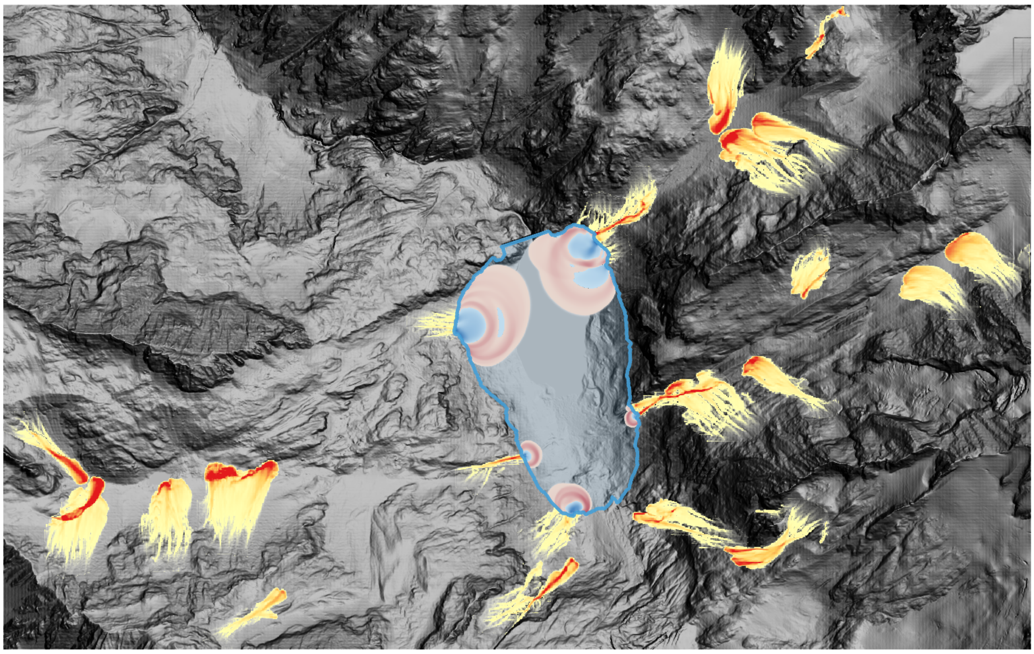


Figure 19: Recommended coupled frame: avalanche depth outside the lake over the diverging water-surface displacement. The semitransparent light-blue polygon and blue border show where the initial lake begins.

18 Profile Analysis accordion

Create or load a QGIS line layer. A convenient method is to create a temporary scratch line layer in the project CRS, toggle editing, and digitize the desired cross section over the DEM. Save the layer if it must persist with the case. The **Profile feature** list allows a feature to be chosen directly; a manual QGIS selection is not required. Then choose the AVAC or WAVE variable and one of three profile modes:

- the currently selected temporal frame;
- the complete temporal series, exportable as CSV or PNG frames; or
- the historical maximum at every sampled point along the line.

Select **Extract / Plot Profile** for an interactive plot and **Export Profile CSV** for sampled distance and values. WAVE water-surface profiles show terrain, water fill, and water elevation; the same profile style is used consistently for compatible AVAC quantities.

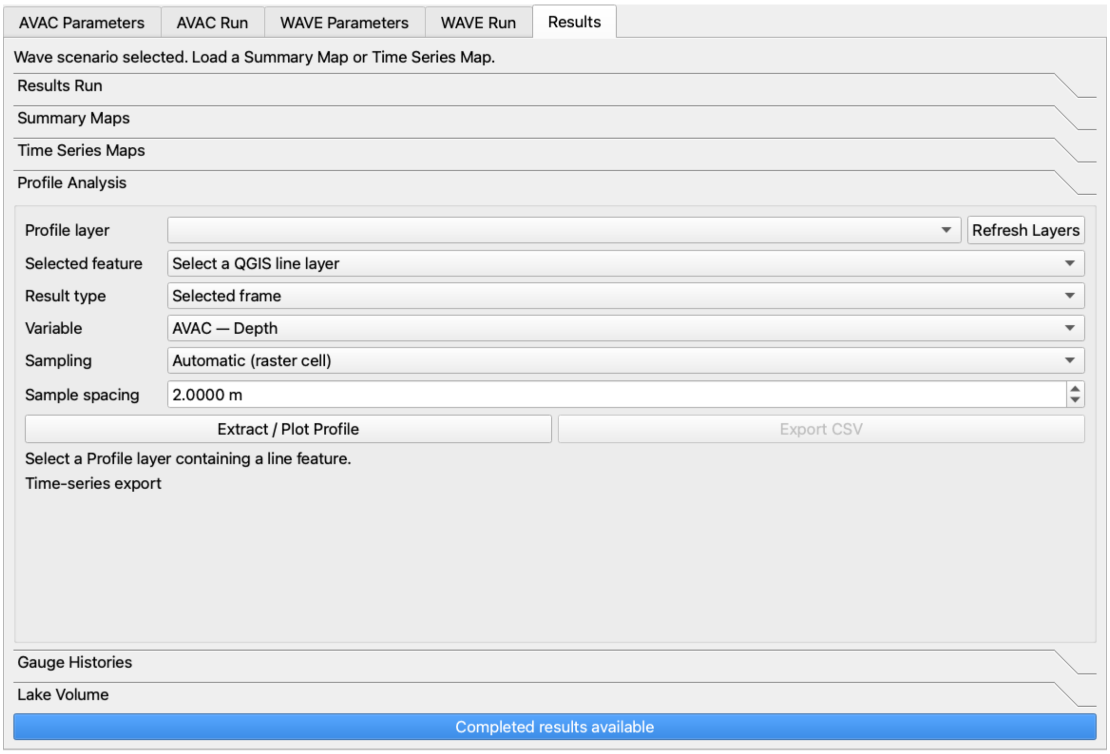
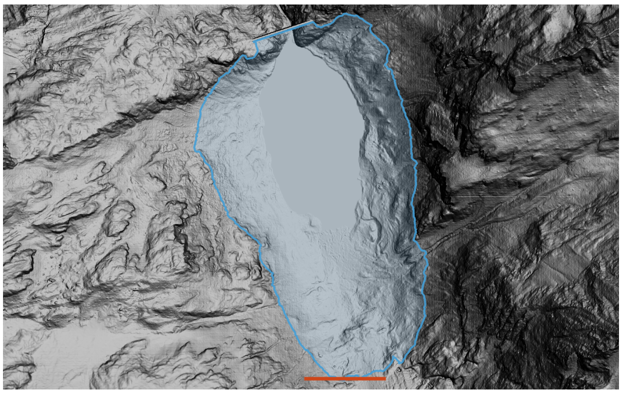
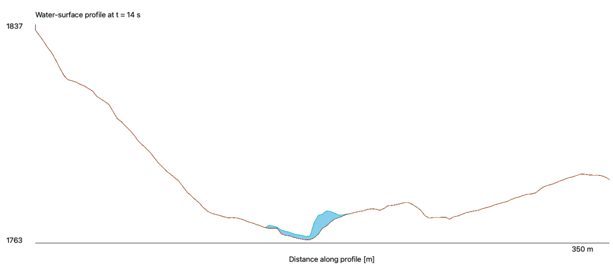


Figure 20: Profile Analysis accordion. Result source, variable, frame mode, line layer, and feature are chosen in one place.



(a) Profile line (orange) digitized over the terrain and lake.



(b) Extracted terrain and water-surface cross section.

Figure 21: Profile workflow from its QGIS line feature to the resulting plot. The brown line is terrain and the blue fill is the water column.

19 Gauge Histories accordion

Gauge histories are postprocessing samples. They do not have to be defined before AVAC or WAVE runs. Load a point layer or create a temporary scratch point layer in QGIS, toggle editing, and digitize one or more locations over the DEM. Save the point layer if it must be reused later. Choose an AVAC or WAVE variable and select **Read Gauges from Selected Point Layer**. The plugin lists the usable point features, transforms their coordinates to the result CRS, and samples each point through all frames.

Choose a sampled point to plot its history or export simulation time and value to CSV. If the required temporal product is not loaded, the tool loads it before sampling.

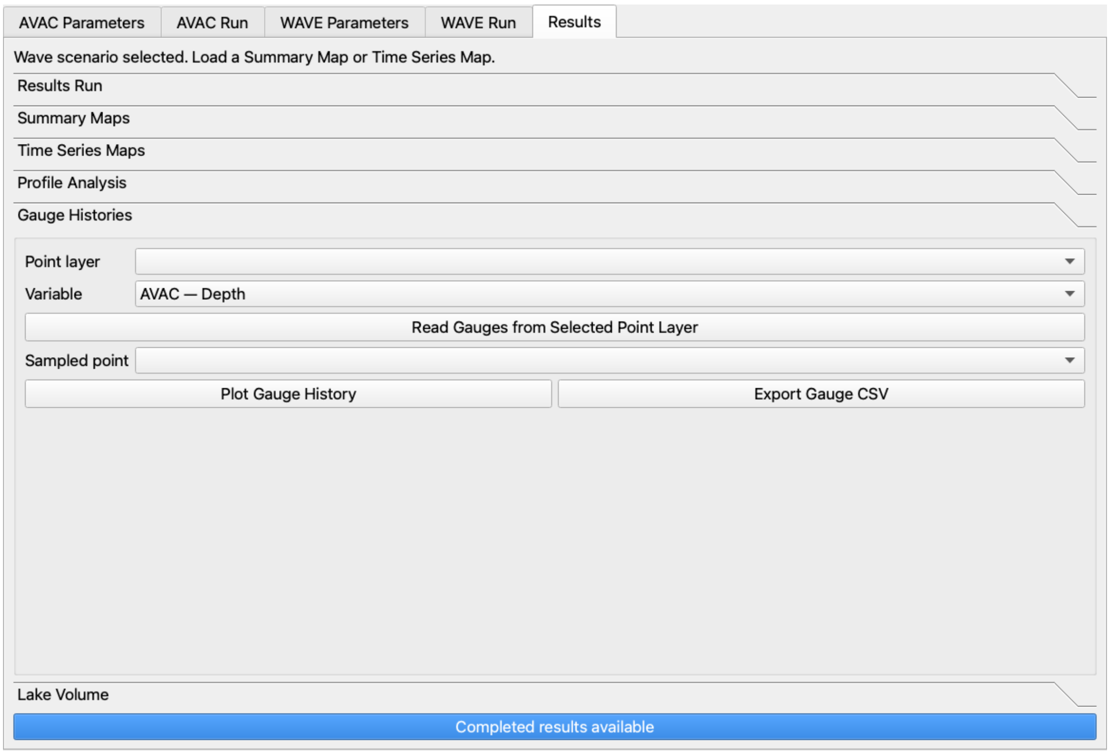
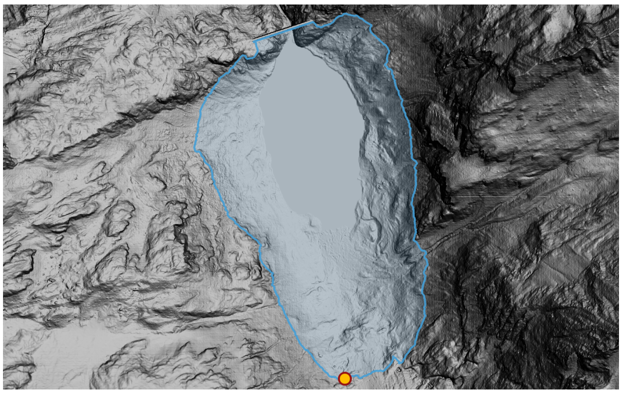
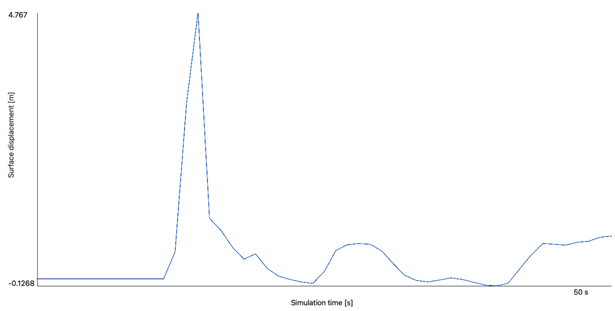


Figure 22: Gauge Histories accordion, shared by avalanche and water variables.



(a) Gauge point (gold marker) placed on the QGIS map.



(b) Sampled water-surface-displacement history.

Figure 23: Gauge workflow from a QGIS point feature to its time-series plot. Later oscillations remain visible after the first arrival.

20 Lake Volume accordion

This final accordion is visible only when WAVE is enabled. It plots or exports the lake-water volume at every WAVE output time. Volume is calculated from water depth and cell area over the initial water-body mask. It is a useful mass-history diagnostic; the current interface does not provide an outflow or spillway hydrograph.

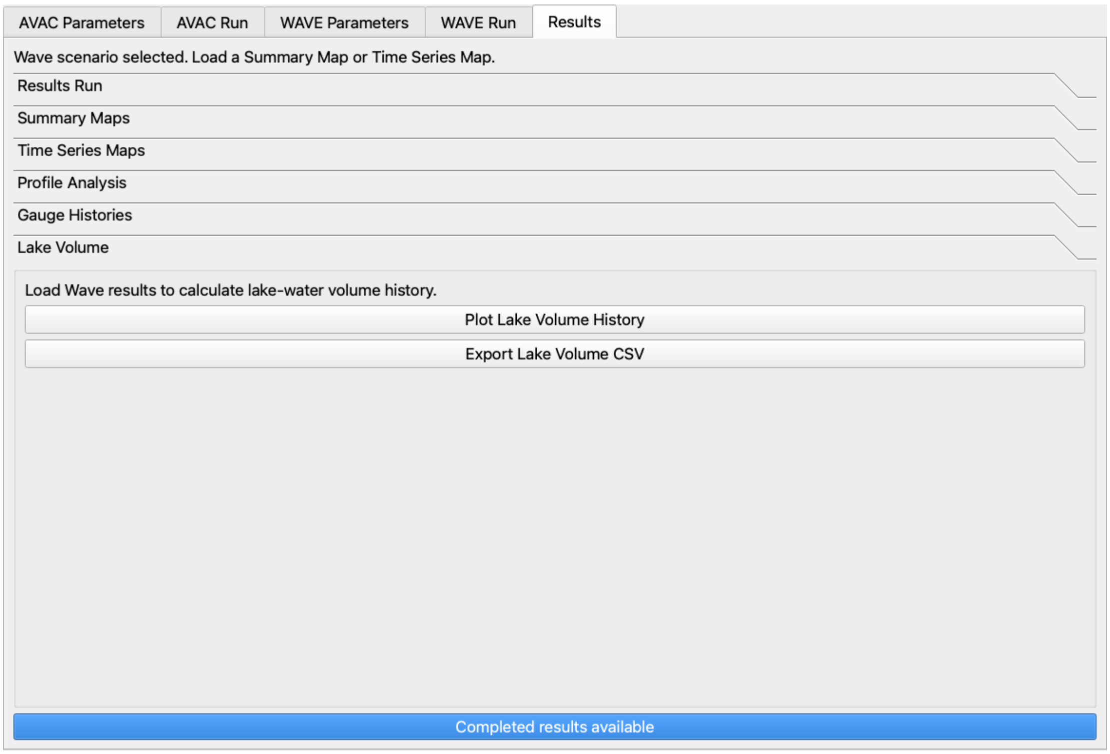


Figure 24: WAVE-only Lake Volume accordion at the end of the Results tab.

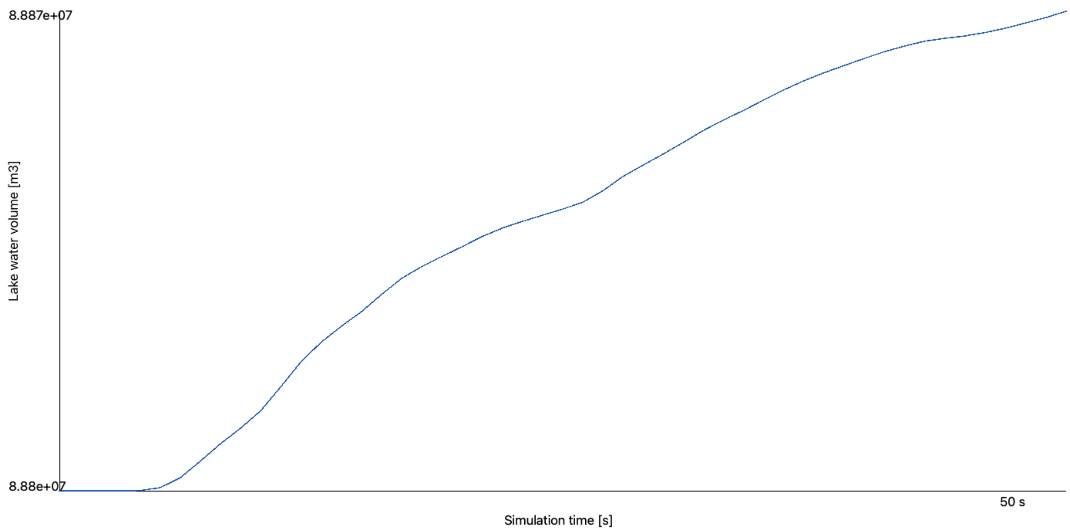


Figure 25: Example lake-water-volume history for a completed coupled simulation.

21 Completion checklist

Before archiving or sharing a case, confirm that:

- the complete setup has been saved with **Save Case**;
- the intended AVAC run and, when applicable, WAVE scenario show a completed status;
- release depth, rheology zones, and initial lake water have been visually checked;
- the WAVE coupling ledger contains plausible transferred mass and both momentum components;
- summary maps and the complete temporal sequence have been inspected, not only a single attractive frame; and
- exported figures and CSV files are stored in the selected AVAC Working Directory with enough metadata to identify their run.

For a control-by-control description, units, persistence rules, and file structure, consult the companion *AVAC4QGIS User Interface Guide* in `docs/ui_reference/`.